



The demand is so high that when one year is left before graduation, each student has at least two offers in hand from highly reputed

organisations — Prof. V. Ramgopal Rao, Department of Electrical Engineering, IIT Bombay



tions,” informs Prof. V. Ramgopal Rao, Department of Electrical Engineering, IIT Bombay.

To address the shortage of engineers possessing the right skills, the Indian government has set up a special manpower development programme (SMDP) in this area and spent a huge amount to introduce VLSI in 25 institutes like NITs and IITs.

Roles you could choose

Chip designers are basically design engineers who define the chip's architecture, create circuit designs, run simulations, supervise layout, tape out the chip to the foundry and evaluate the prototype once the chip comes back from the laboratory.

At the top of the ladder is the architectural team, which designs the actual chip. The defined design is then implemented in a high-level language by the logic design team. Verilog and VHDL are the two popular languages being used today to write the code. With the new paradigm of system-on-chip design, SystemC and SystemVerilog are being used for top-level design.

The circuit design team synthesises the logic description into a circuit, producing a document referred to as 'netlist.' The physical design team then takes the netlist and performs a 'layout' of the chip.

It's the job of an application engineer to make sure that the chip works in the system that has been designed or used by customers. Process engineers, on the other hand, are involved in new wafer process development, device modeling and other R&D projects. Packaging engineers develop precision

packaging technology and new package designs for chips.

This is followed by testing and verification. For every two design engineers, there are five verification engineers. Freshers are entrusted with the job of testing a chip, while its designing is done only by experienced professionals, who ensure (at all stages) that the design bestows the chip with following attributes: Low power consumption, less noise, less area usage, and higher efficiency in terms of speed and yield.

Jobs are primarily available in digital design, analogue design, electronic design automation, embedded software, and product and test engineering. There is a strong demand for talented system engineers and application engineers. Depending on whether a candidate is a fresher or experienced professional, his role in the organisation could range from a technical individual-contributor to a technical leader to an engineering manager.

How to get the entry?

A bachelor's degree in electronics, communications or instrumentation is a prerequisite. Companies look for candidates with knowledge of the circuit design, VLSI design, microcontrollers, IC design, C programming, Assembly-level programming, etc.

There are many universities in

Institute Watch

1. IITs (Delhi, Guwahati, Mumbai, Madras, Kanpur, Kharagpur, Roorkee)
2. VEDA IIT, Hyderabad
3. Centre for Development of Advanced Computing (C-DAC)
4. IISc, Bangalore
5. BITS, Pilani
6. University of Pune
7. NIT, Tiruchirappalli
8. NIT Rourkela
9. Netaji Subhas Institute of Technology (NSIT)
10. Manipal Centre for Information Science
11. Punjab University, Chandigarh
12. IIT, Hyderabad
13. Delhi University
14. Anna University
15. VJTI Mumbai
16. PSG College of Engineering, Coimbatore
17. Jadhavpur University

India that offer good education in this field. All IITs, NITs and other Tier I and II colleges offer high-quality education in chip designing. Some of the courses offered at these institutes cover microelectronics, microprocessors, VLSI design, and digital and analogue systems.

“We work closely with many universities in the country and help them with setting up of labs so that students get opportunities to work on practical aspects as part of their coursework. We also help universities to periodically revise their syllabus to ensure that students are studying the most updated technology and the latest breakthroughs during their study years,” says Sanjay Bhan, director-HR, Texas Instruments India.

Wipro Technologies primarily looks for graduates in engineering and



The salary across the market is very good for experienced candidates. It is also equally rewarding for freshers in terms of exposure and growth — Teresa Mishra, general manager-talent acquisition, Wipro Technologies